Han Zhou

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Education

University of Waterloo — Waterloo, ON

BMath Computer Science (Honours with Co-op) & Combinatorics and Optimization — *Sept. 2023 – May. 2028*

- GPA: 3.9/4.0

Technical Skills

Languages/Frameworks/Tools: Python, C/C++, Java, Rust, PostgreSQL, PyTest, PyTorch, Transformers, VLLM, NumPy, WandB, Slurm, Docker/Apptainer, GCP, Git, Linux, Shell, Matlab

Experience

Noah's Ark Lab — Montreal, QC

LLM Reasoning Researcher — *Jan. 2026 – Aug. 2026*

- Architected and implemented a statistical testing framework for **LLM chain-of-thought reasoning** in **Pytorch** and **Transformers**, supporting interchangeable **test-time computation** backends;
- Reduced memory usage of **beam-search** baselines for LLM **probabilistic commonsense abduction**, enabling evaluation on V100 GPUs on diverse datasets; paper accepted by ICML 2026;
- Repurposed latent-skill baselines for agents with **Apptainer**, **VLLM**, **Slurm** for different experimental protocols.

University of Waterloo Combinatorics & Optimization Department — Waterloo, ON

Undergraduate Research Assistant under Prof. M. Meila, Prof. S. Vavasis, Prof. X.V. Doan — *Jan. 2026 – Present*

- Created **Matlab** experiments with **cvx** to explore novel **subspace-clustering** relaxations;
- Optimized a **convex optimization** implementation to reduce memory usage on large-scale datasets.

Sibli — Toronto, ON

ML Software Engineer — *May. 2025 - Aug. 2025*

- Fine-tuned **spaCy** models for **Named Entity Recognition (NER)**, elevating the **f1-score** from **31%** to **44%** and enabling more accurate and reliable automated extraction of organizations from documents;
- Further improved the NER performance by experimenting with **BERT** and **ModernBERT** models, implemented using **Hugging Face** and **Pytorch** with **partial freezing**, raised precision to **78%** and recall to **84%**;
- Developed a **fuzzy search** algorithm to process outputs from **Gemini Vertex and ChatGPT agents**, created a **Docker** container to run on **Google Cloud Run**, successfully constructing a corporate knowledge graph.

EclipseCard Inc. — Waterloo, ON

Software Engineer Intern — *May 2024 – Aug. 2024*

- Built scalable **Python/Flask REST APIs** with **PyTest** coverage for personalized credit-card recommendations.
- Implemented **Metropolis-Hastings** in **Rust**, and deployed a Python **KNN** model to forecast user savings.

Projects

TinyNTK | *Python, Numpy, Pytest* — Aug. 2026

- Built a **NumPy-only** reverse-mode **autodiff** engine with VJP-based backpropagation, aimed at exploring empirical **Neural-Tangent Kernels (NTKs)** for vector-valued networks with layerwise decomposition.
- Verified gradients and kernels using finite differences and analytical NTK formulas.

TRM Reimplementation | *Python, Pytorch, FlashAttention, Slurm* — Apr. 2026

- Reimplemented a **Tiny Recursive Model (TRM)** with recursive reasoning and deep supervision;
- Built a GPU/Slurm training pipeline with checkpointing, evaluation, and reproducible experiment workflows.

Mini GPT2 | *Python, PyTorch, Transformers* — May. 2025

- Implemented a GPT-2-style Transformer decoder from scratch in PyTorch, including **12 Transformer blocks**, causal **multi-head self-attention**, MLPs, LayerNorm, positional embeddings, and unembedding; trained a smaller 2-layer Transformer on Neel Nanda's Pile-10k dataset with next-token cross-entropy and AdamW.

Vm Editor | *C++, NCurses, Git* — Dec. 2024

- Built an **MVC-based** terminal editor with low-coupling architecture, Vim-style commands, undo/redo, search.